

Annamaria Letizia Palmiero

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Education

University of Maryland, Baltimore County (UMBC)
B.S. Statistics; B.S. Computer Science. Minor: Mathematics.

Expected May 2026
CGPA: 3.80 / 4.0

Experience

Mathematics Research Assistant, Yale University

Jun 2025–Present

- Conduct research in computational fluid dynamics focusing on matrix decomposition, dimensionality reduction, and machine-learning prediction of high-speed turbulent flows.
- Co-authoring manuscript on decomposition techniques for characterizing vortex structures.

Student Intern, Department of the U.S. Navy

Jan 2024–Present

- Designed and deployed a Power Apps–SharePoint system with automated Power Automate workflows as the sole developer.
- Standardized and validated large multi-platform datasets (SQL, Tableau, data dictionaries) to ensure reproducibility and consistency.
- Led trainings on time-series analysis and data visualization for 50+ personnel.

Research Assistant, UMBC Division of Information Technology

Jan–May 2024

- Maintained predictive and inferential models (machine learning, logistic/linear regression) for student success analytics and enrollment forecasting.
- Performed statistical analyses, validated data pipelines, and visualized trends to inform decision-making and optimize predictive models.

Self-Directed Researcher, UMBC

Jan–May 2023

- Developed supplemental programming challenges and quantitatively analyzed their effect on student performance in an introductory computing course.
- Conducted statistical analysis in R, and presented findings at UMBC’s Teaching & Learning Symposium.

Publications

Mai Al Shaaban, Joey Takei, **Annamaria Palmiero**, Leya Dereje. “Investigation of Decomposition Techniques for Characterizing Complex Vortex Structures in MVG Controlled Boundary Layer.” Planned submission to *Algorithms* (MDPI).

Conference Presentations

Oral Presentations

- **Matrix Decomposition for Characterizing Vortex Structures**
Young Mathematicians Conference, Ohio State University (Aug 2025).
- **Matrix Decomposition in Computational Fluid Dynamics**
Yale University–Williams College Undergraduate Mathematics Research Conference (Jul 2025).

- **Exploring Research with R: From Data to Insight**
STEM Student Leadership Conference, Loyola University (May 2025).
- **Visualizations in Processing**
UMBC Programming Lecture Series (Apr 2025).

Poster Presentations

- **Matrix Decomposition for Complex Vortex Structure Characterization**
Joint Mathematics Meeting, Washington D.C. (to be presented Jan 2026).
- **Machine Learning–Based Prediction for Vortex Structures in High–Speed Flows**
MIT Undergraduate Research Technology Conference (Oct 2025).
- **Matrix Decomposition & Prediction for Vortex Structure Characterization**
UMBC Summer Undergraduate Research Festival (Aug 2025).
- **Machine Learning–Based Vortex Evolution**
Jane Street Summer Mathematics Day (Jul 2025).
- **The Effects of External Learning on Computer Science Performance**
UMBC Teaching & Learning Symposium (Apr 2023).

Grants, Fellowships & Honors

- National Science Foundation Travel Grant, \$500 — Young Mathematicians Conference (2025)
- Whiting-Turner Research Grant, \$2,000 — Yale University Research (2025)
- UMBC Mathematics Department Research Grant, \$2,000 — Yale University Research (2025)
- France-Merrick Fellowship, \$8,500 – Commitment to Service and Leadership at UMBC (2023–2024)
- Phi Beta Kappa (Inducted 2025); Pi Mu Epsilon (Inducted 2025); Mu Sigma Rho (Inducted 2024)
- College of Natural and Mathematical Sciences Scholar (2023–Present)
- President’s List (Fall 2024, Spring 2025); Dean’s List (Fall 2023)

Teaching Experience

- **Lead Teaching Fellow**, Computational Thinking and Design — Aug 2024–Present
- **Teaching Assistant**, Multivariable Calculus — Aug 2023–Dec 2023
- **Teaching Fellow**, Computational Thinking and Design — Aug 2022–May 2023

Leadership

- **President**, Mu Sigma Rho Statistics Honors Society (2025–Present)
- **Vice President**, Mu Sigma Rho (2024–2025)
- **Service Site Leader**, Baltimore Animal Rescue and Care Shelter (2024–2025)
- **Treasurer & Volunteer Coordinator**, Animal Welfare Student Organization (2023–2025)
- **Living Learning Community Peer Leader & Mentor** (2022–2025)

Skills

- **Statistical:** Probability theory, statistical inference, regression theory, multivariate statistics, statistical computing, stochastic processes, data cleaning, visualization.
- **Computational:** Machine learning, algorithm analysis, data structures, automata theory, efficient coding practices, object-oriented design.
- **Programming:** R, Python, C++, C, L^AT_EX, Assembly, MATLAB, Processing.
- **Software:** Power Apps, Power Automate, Power BI, Tableau, Excel, Word, PowerPoint.